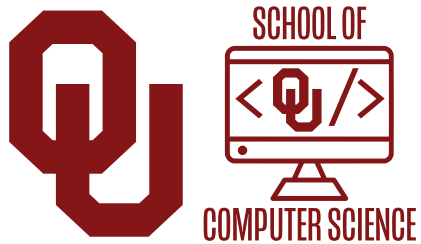
Evolutionary Lattice Sieves

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Doctoral Dissertation Defence, 2026





Outline

1. Introduction
   - Computer Science
   - Motivation
2. Preliminaries
   - Quantum Computers
   - The Knapsack Problem
   - Lattices
3. Approaches
   - Traditional Approaches
   - Our Domain-Informed Genetic Algorithm (GA)
4. Results
   - Test Data
   - Reduction Performance
5. Conclusion
   - Summary & Outlook



Computer Science

What are the relevant disciplines?

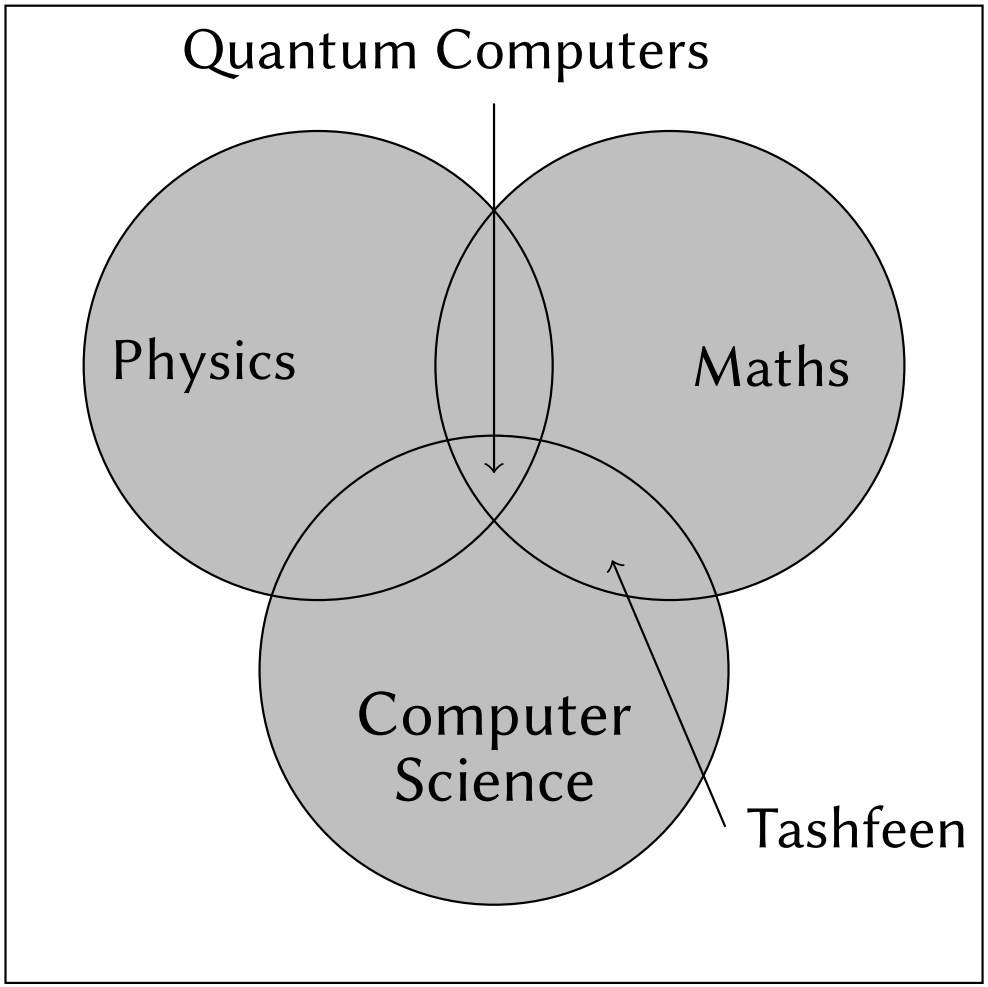


Figure: The overlap of quantum computers across fields of study.



Computer Science

Is it dead? No.



Every computer program is a model, hatched in the mind, of a real or mental process. These processes, arising from human experience and thought, are huge in number, intricate in detail, and at any time only partially understood. They are modeled to our permanent satisfaction rarely by our computer programs. Thus even though our programs are carefully handcrafted discrete collections of symbols, mosaics of interlocking functions, they continually evolve: we change them as our perception of the model deepens, enlarges, generalizes until the model ultimately attains a metastable place within still another model with which we struggle. The source of the exhilaration associated with computer programming is the continual unfolding within the mind and on the computer of mechanisms expressed as programs and the explosion of perception they generate. If art interprets our dreams, the computer executes them in the guise of programs!

—Structure and Interpretation of Computer Programs (SICP), 2nd ed. (pg. xiii-xiv)

Computer Science

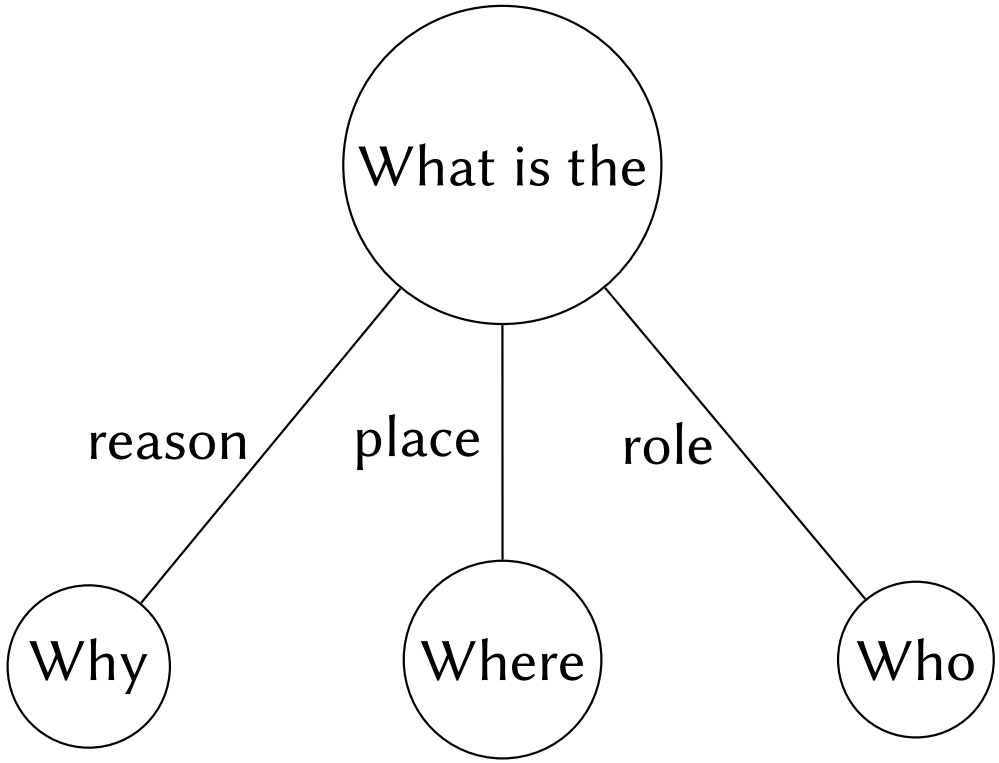
Is Computer Science just Mathematics with a moustache?

What : Mathematics

Why : Philosophy

Where : Geography

Who : Political Science?



- Does there exist an x for $x \in \{\text{reason, place, role}\}$?
- If it does, what is the best "method" to find x: *How* to find x?
- Mathematics provides a framework for dealing precisely with notions of "what is." Computation provides a framework for dealing precisely with notions of "how to." —SICP (pg. xxiii)



Feynman on—what differs physics from mathematics?
https://youtu.be/B-eh2SD54fM



Example (Ways to order a 12 doughnuts at Dunkin')

- Whether arrangement matters (or not) and yes (no) repetitions.

$$x_1 + x_2 + x_3 + \cdots x_{92} = 12$$

x_i corresponds to the number of doughnuts we order of the ith type.

$$\binom{12 + 92 - 1}{12} = \frac{103!}{12!(103 - 12)!} = 1,529,143,625,215,220$$

- This is more than the number of estimated stars in the Milky Way.
- How do we generate/enumerate these 1,529,143,625,215,220 orders?



Example (What are the least amount of coins that allow to make any change?)

Change A sub-dollar amount 1p to 99p.

Amount of coins Amount(s) of quarters, dimes, nickels and pennies.

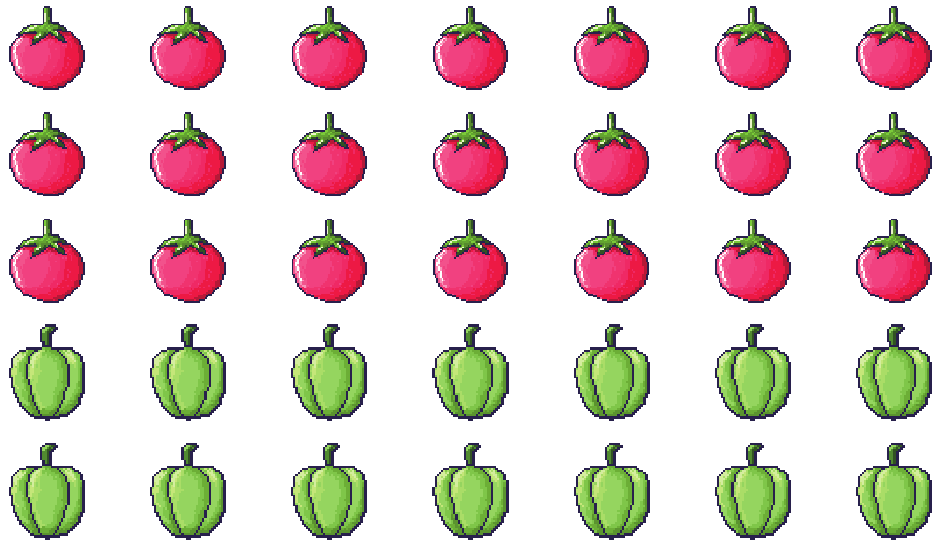
Intermediate Value Theorem If Alice weighed 150 pounds on a Monday and 148 on the following Sunday then there has to be a moment/time in-between where Alice weighed 149 pounds.

Algorithm Try to make 99p.



Example (Optimal Crop Layout)

If you have 21 tomato plants and 14 peppers, how can you arrange them in a plot such that it is rectangular and no rows have two types of vegetables? Each row can be $\gcd(21, 14)$ vegetables long. Since, $14 = 2 \times 7$ and $21 = 3 \times 7$, we may inspect that $\gcd(21, 14) = 7$.



Computer Science

How about the fashionable abuse of notation?

Yi'all still write this $f(n) = O(g(n))$, and mean this $f(n) \in O(g(n))$.

Non-Deterministic Polynomial Time (Class NP) are problems where solution can verified in Deterministic Polynomial Time (Class P) but we don't know whether it can be found in P-time. We colloquially refer to P problems as easy and NP as difficult. A better terminology maybe understood and not-understood since $O(n^{2^{100}})$ is still P-time.



Motivation

What is Cryptography?

CRYP·TOG·RA·PHY *(krip-TOG-ruh-fee)* ***n.***

[Gr. *κρυπτός* (kryptos) hidden + *γράφειν* (graphein) to write]

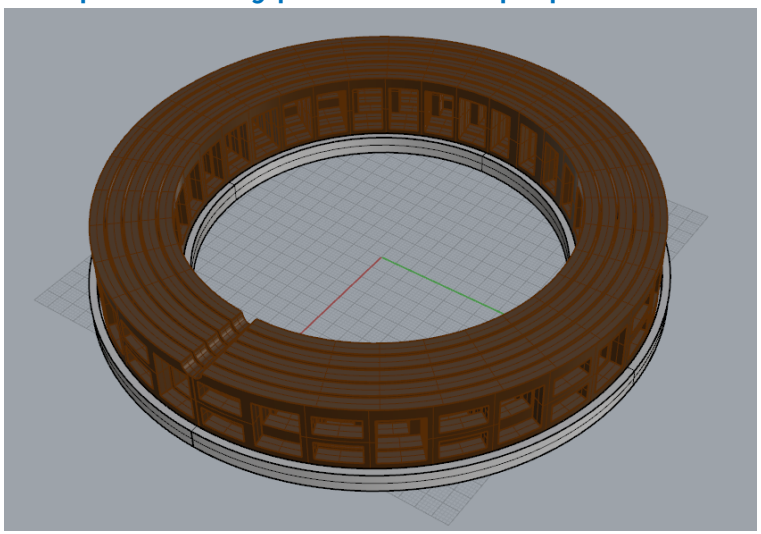
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1 The art and science of writing or solving codes.

2 The branch of mathematics and computer science concerned with the principles and techniques of secure communication in the presence of adversaries, encompassing encryption, decryption, and authentication.

– **cryp·tog·ra·pher**, *n.* – **cryp·to·graph·ic**, *adj.*

https://cryptii.com/pipes/caesar-cipher/





Motivation

Why do hard problems matter and their consequences in Cryptography?

Alice
Represents message as a seemingly hard problem p using special information

$\xrightarrow[p \text{ sent over a public channel}]{\text{Eve to solve } p \text{ without special info.}}$

Bob
Bob solves p for message using the special information

Take RSA for example,

Alice
Encrypts m as $m^e \bmod N$ and inverting $e \bmod \varphi(N)$ is hard without special information $N = p \cdot q$

$\xrightarrow[\text{where } \varphi(N)=(p-1)(q-1)]{\text{Eve to solve for } d=e^{-1} \bmod \varphi(N)}$

Bob
Bob knows p, q therefore, can $m^{de} = m \bmod \varphi(N)$



Computational Hardness

Hardness due to computational intractability, i. e., security of the system is by the computational bottle neck of the time.

- Along with the integer factorisation (RSA), another classical hard problem is the discrete log (Diffie-Hellman) etc.
- Factoring is in NP and there is no proof that there does not exist a P solution.
- Recall that Vigenère cipher was broken after three centuries of no solutions [4].



Quantum Computers

What is the difference between a classical and a quantum computer?

- We know a classical bit can be in one of the two states,

$$\psi \in \{0, 1\}, \quad P(\psi = 1) = P(\psi = 2) = \frac{1}{2}$$

- A classical bit is a special case of a quantum bit (qubit) $|\psi\rangle$,

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \qquad \alpha, \beta \in \mathbb{C}$$

$$= \begin{pmatrix} \alpha \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \beta \end{pmatrix}$$

$$= \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \alpha|0\rangle + \beta|1\rangle \qquad |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$



Born's Rule

The probability that a quantum state $|\psi\rangle$ will collapse into one of the possible classical states $|0\rangle, |1\rangle$ upon observation is, e. g.,

$$P(|\psi\rangle = |0\rangle) = ||\alpha||^2 = |a + bi| \cdot |a - bi| = a^2 + b^2$$

- Then the qubit is more precisely defined as,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad ||\alpha||^2 + ||\beta||^2 = 1 \quad \text{Hilbert Space over } \mathbb{C}^2$$

- The Schrodinger's Cat constants: $\alpha = \beta = \frac{1}{\sqrt{2}}$ satisfy $||\alpha||^2 + ||\beta||^2 = 1$.

$$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = 1$$

- Honestly, Quantum Mechanics feels like a maths meme physicists took too seriously.



Quantum Computers

What is the threat (if there ever will be any)?

- Some classically hard problems such as the ones mentioned previously (integer factorisation and discrete log) are easy to solve by running Shor's algorithm on a quantum computer armed fancy qubits.
- Qubits are hard to build without too much noise.
- Encrypted data stored today can be decrypted in future.



The Knapsack Problem

What are the problems for which there is no known polynomial time quantum algorithm?

We define the *knapsack* problem as, for any $M \in \mathbb{N}^n$, $S \in N$ find $x \in \{0, 1\}^n$ such that,

$$M \cdot x = S$$

From the changing making example,

$$M' = [1, 5, 10, 25], x = [1, 1, 0, 1]$$

Then,

$$M' \cdot x = 31 \text{ cents.}$$

If any of the NP-complete problems are shown to be in P then all NP-problems are in P. In the same seminal paper where Karp determines a subset of the NP problems as NP-complete with the aforementioned property, he also gives twenty-one examples of such NP-complete problems. Number 18 on the list is the knapsack or the subset-sum problem.



The Knapsack Problem

When is it easy?

Super Increasing Sequences These are sets $M' = \{r'_1, r'_2, r'_3, \cdots, r'_n\}$ with $r'_{i+1} \geq 2r'_i$. What's an example? You've seen one.

Coin Denominations $M' = \{1, 5, 10, 25\}$

Input: Superincreasing set M_{easy} and integer sum S.
Output: Solution to the knapsack problem x.

1. **for** $i \in \{|M_{\text{easy}}|, |M_{\text{easy}}| - 1, |M_{\text{easy}}| - 2, \ldots, 1\}$
2. $\quad x_i \leftarrow 0$
3. $\quad$ **if** $S \geq (M_{\text{easy}})_i$
4. $\quad\quad x_i \leftarrow 1$
5. $\quad\quad S \leftarrow S - (M_{\text{easy}})_i$

Algorithm: Linear time algorithm to solve the knapsack problem for superincreasing sets M_{easy}.



The Knapsack Problem

How can it be used to build a cryptosystem?

The Merkle-Hellman scheme is as follows,

Alice	Eve	Bob
Pick $M' = [r'_1, \ldots, r'_n]$, such that $r'_1 > 2^n$, $r'_{i+1} \geq 2r'_i$. Pick A, B with $B > 2r'_n$ and $\gcd(A, B) = 1$.		



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	M	
Let $r_i \equiv Ar'_i \mod B$ & $M = \{r'_i \in M' : r_i\}$ $\longleftarrow$		$\longrightarrow M$



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	S	
S		$S = Mx$
(M', S') is $\mathcal{O}(n)$.	(M, S) is $\mathcal{NP}$-imposter.	



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	S ⟵	⟵ $S = Mx$
(M', S') is $\mathcal{O}(n)$. Let $S' \equiv A^{-1}S \mod B$. Solve $(M', S') \to x$. We have $M'x = S'$.	(M, S) is $\mathcal{NP}$-imposter.	



The Knapsack Problem

Why is $M'x = S'$ if and only if $S = Mx$?

$$
\begin{aligned}
S' &\equiv A^{-1}S \mod B \\
&\equiv A^{-1}Mx \mod B && \text{Bob's Encryption } S = Mx \\
&\equiv \sum_{i=1}^{n} A^{-1} r_i x_i \mod B && \text{Since } M = \{r'_i \in M' : r_i\} \\
&\equiv \sum_{i=1}^{n} A^{-1}(Ar'_i) x_i \mod B && \text{Since } r_i \equiv Ar'_i \\
&\equiv \sum_{i=1}^{n} r'_i x_i \mod B \\
&\equiv M'x \mod B \\
&= M'x && \text{Since } M'x \leq r'_1 + r'_2 + r'_3 +, \ldots, r'_n < 2r'_n < B
\end{aligned}
$$



The Knapsack Problem

Why can any general knapsack problem be solved in $O(2^{|M|/2})$?

We show an example where $M = \{2, 3, 5, 7, 11, 13\}$ and $S = 26$,

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} & 0 \\ & 2 \\ & 3 \\ & 5 \\ & 5 \\ & 7 \\ \ell \longrightarrow & 8 \\ \ell \longrightarrow & 10 \end{bmatrix} \star \begin{bmatrix} 0 & \leftarrow r \\ 7 & \leftarrow r \\ 11 & \leftarrow r \\ 13 & \leftarrow r \\ 18 & \leftarrow r \\ 20 & \\ 24 & \\ 31 & \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \\ 13 \end{bmatrix}$$

$$L_{\ell,1} + R_{r,1} \in \{10, 17, 21, 23, 28, 26\}$$

We recovered $x = [0, 1, 1, 1, 1, 0]$ in less than $2^{6/2} = 8$ steps.



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Lattices

What is an integral lattice?

- A lattice in 2D is a grid of points, i.e., integral linear combinations.
- Consider a basis $\mathcal{B} \in \mathbb{R}^{d \times d}$ with d linearly independent vectors,

$$\text{Integral Span of } \mathcal{B} = \mathcal{L}(b_1, \ldots, b_d) = \{x_1 b_1 + x_2 b_2 + \cdots + x_d b_d : x_i \in \mathbb{Z}\}$$

$$\mathcal{L}(\mathcal{B}) = \{\mathcal{B}x : x \in \mathbb{Z}^d\}$$

$$\dim(\mathcal{L}) = d$$

$$\det \mathcal{L}(\mathcal{B}) = |\det \mathcal{B}|$$



- Define *Gaussian integers* as,

$$\mathbb{Z}[i] = \{\alpha + \beta i : (\alpha, \beta) \in \mathbb{Z}^2\}, \quad i^2 = -1$$

  Despite the name, Gaussian integers are complex numbers $\alpha + \beta i$ where both the real $\Re(\alpha + \beta i) = \alpha$ and the imaginary $\Im(\alpha + \beta i) = \beta$ parts are limited to the integers.
- Form a lattice, but this time in the ambient space of complex numbers $\mathbb{C}$ with $\mathcal{B} \in \mathbb{Z}[i]^{d \times d}$

$$\mathcal{L} = \{\mathcal{B}x : x \in \mathbb{Z}[i]^d\}$$

- The norm of a $v \in \mathbb{Z}[i]^d$ is given as,

$$\|v\|_2^2 = v^H v = |v_1|^2 + |v_2|^2 + \cdots + |v_d|^2$$

  where $|v_j|^2 = \alpha^2 + \beta^2$ for $v_j = \alpha + \beta i$. The v^H is known as the *conjugate transpose* of v.



Lattices

What are good and bad bases?

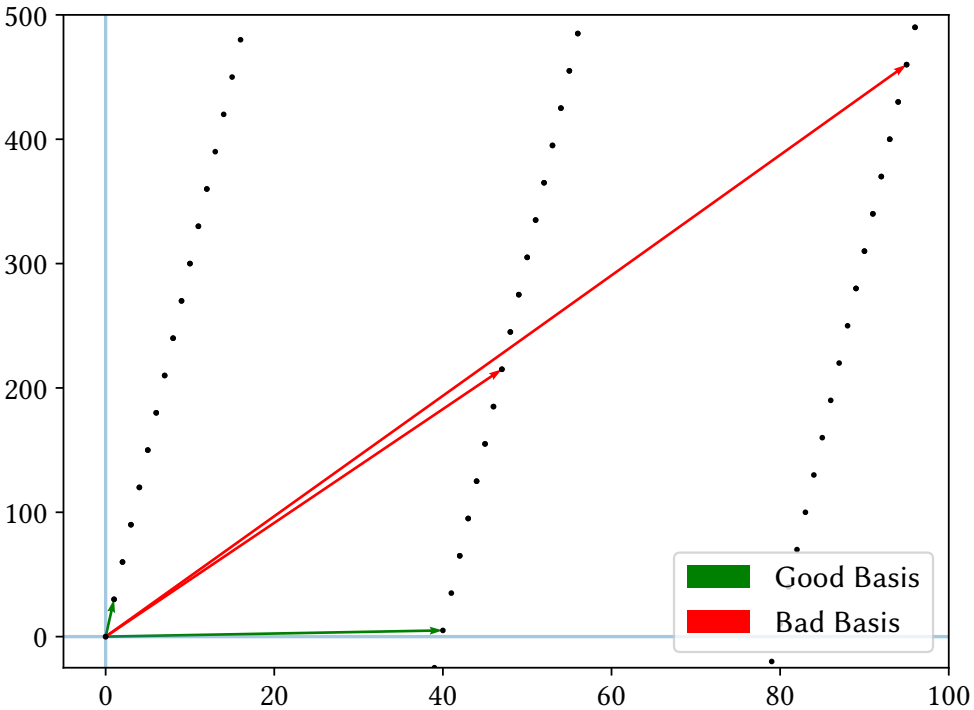


FIGURE: Example of a 2D lattice.

$$\mathcal{B}_{\text{bad}} = \begin{bmatrix} 95 & 47 \\ 460 & 215 \end{bmatrix}$$

$$\mathcal{B}_{\text{good}} = \begin{bmatrix} 1 & 40 \\ 30 & 5 \end{bmatrix}$$

$$C = \begin{bmatrix} 15 & 7 \\ 2 & 1 \end{bmatrix}$$

$$\mathcal{B}_{\text{bad}} = \mathcal{B}_{\text{good}} C$$

$$\det C = 1$$

- SVP: Find the shortest vector in $v_{\text{shortest}} \in \mathcal{L}(\mathcal{B})$. It is $\mathcal{NP}$-hard.
- apprSVP: Find a vector that is at most α times the length of v_{shortest}.



Lattices

How can lattices be used to solve the knapsack problem?

Take any knapsack problem (M, S) and the relevant solution x such that $Mx = S$. Now consider the lattice basis $\mathcal{B}_{\text{bad}}$ of dimension $d = n + 1$. The lattice spanned by $\mathcal{B}_{\text{bad}}$ must have a vector $t \in \mathcal{L}(\mathcal{B}_{\text{bad}})$ that is the result of an integral linear combination due to x.

$$t = \underbrace{\begin{bmatrix} 2 & 0 & \cdots & 0 & 1 \\ 0 & 2 & \cdots & 0 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 2 & 1 \\ r_1 & r_2 & \cdots & r_n & S \end{bmatrix}}_{\mathcal{B}_{\text{bad}}} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ -1 \end{bmatrix} = \begin{bmatrix} 2x_1 - 1 \\ 2x_2 - 1 \\ \vdots \\ 2x_n - 1 \\ M \cdot x - S \end{bmatrix} = \begin{bmatrix} 2x_1 - 1 \\ 2x_2 - 1 \\ \vdots \\ 2x_n - 1 \\ 0 \end{bmatrix}$$

$$x \in \{0, 1\}^n \Rightarrow 2x_i - 1 = \pm 1 \Rightarrow ||t|| = \sqrt{n}$$

$||t||$ is at a stark contrast with the other vectors in the lattice spanned by $\mathcal{B}_{\text{bad}}$ due to the relative size of $m_i \in \Theta(2^{2n})$.



Eve has,

$$
\begin{aligned}
S &= 2002491457667039 \\
M &= [r_1, r_2, r_3, \cdots, r_{25}] \\
&= [67108861, 134217725, 268435453, 536870909, 1073741821, 2147483645, \\
&\quad 4294967293, 8589934589, 17179869181, 34359738365, 68719476733, 137438953469, \\
&\quad 274877906941, 549755813885, 1099511627773, 2199023255549, 4398046511101, \\
&\quad 8796093022205, 17592186044413, 35184372088829, 70368744177661, \\
&\quad 140737488355325, 281474976710653, 562949953421309, 1125899906842621]
\end{aligned}
$$

And the encoding table,

␣	A	B	$\cdots$	T	Y	Z
0	1	2	$\cdots$	20	25	26
00000	00001	00010	$\cdots$	10100	11001	11010



Eve generates the bad basis.

$$\begin{bmatrix}
2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 \\
r_1 & r_2 & r_3 & r_4 & r_5 & r_6 & r_7 & r_8 & r_9 & r_{10} & r_{11} & r_{12} & r_{13} & r_{14} & r_{15} & r_{16} & r_{17} & r_{18} & r_{19} & r_{20} & r_{21} & r_{22} & r_{23} & r_{24} & r_{25} & S
\end{bmatrix}$$



Eve reduces the bad basis into a good basis.

$$\begin{bmatrix}
-4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 1174258 \\
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0 & 2 & -4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1174260 \\
0 & 0 & 2 & -4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 1174258 \\
0 & 0 & 0 & 2 & -4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 1 & -3 & 0 & 0 & 0 & 0 & 1174262 \\
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0 & 0 & 0 & 0 & 0 & 2 & -4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1174266 \\
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & -4 & 1 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 1175364 \\
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3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 0 & 3 & 3 & 3 & 3 & 0 & 0 & 3 & 3 & 3 & 3 & 782839
\end{bmatrix}$$



Eve reduces the bad basis into a good basis.

$$\begin{bmatrix}
-4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 1174258 \\
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 2 & 3436596 \\
3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 3 & 0 & 3 & 3 & 3 & 3 & 0 & 0 & 3 & 3 & 3 & 3 & 782839
\end{bmatrix}$$



Eve verifies,

$$||t|| = \sqrt{n} = \sqrt{25} = 5$$

She then lets,

$$x \equiv t - 1 \equiv [0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1] \mod 3$$

Breaks x per encoding,

$$00011 \quad 01000 \quad 00101 \quad 01110 \quad 00111$$



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Breaks x per encoding,

00011	01000	00101	01110	00111
↓	↓	↓	↓	↓
3	8	5	14	7



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Breaks x per encoding,

00011	01000	00101	01110	00111
↓	↓	↓	↓	↓
3	8	5	14	7
↓	↓	↓	↓	↓
C	H	E	N	G



Traditional Approaches

How is the GCD a first lattice reduction algorithm?

$$G(a,b) = \begin{cases} G(b,a) & \text{if } \|a\| > \|b\| \\ (a,b) & \text{if } f(a,b) = 0 \\ G(a, b - f(a,b)a) \end{cases}$$

$$G(748, 2024) = G(748, 2024 - (2)748) \qquad \text{Since } f(2024, 748) = \left\lfloor \frac{2024}{748} \right\rfloor = 2$$

$$G(748, 528) = G(528, 748) \qquad \text{Since } \|748\| > \|528\|$$

$$G(528, 748) = G(528, 748 - (1)528) \qquad \text{Since } f(748, 528) = \left\lfloor \frac{748}{528} \right\rfloor = 1$$

$$\vdots$$

$$G(44, 88) = G(44, 88 - (2)44) \qquad \text{Since } f(88, 44) = \left\lfloor \frac{88}{44} \right\rfloor = 2$$

$$G(44, 0) = G(0, 44) \qquad \text{Since } \|44\| > \|0\|$$

Now let $f(b_1, b_2) = \left\lfloor \frac{b_1 \cdot b_2}{b_1 \cdot b_1} \right\rceil$ and open dissertation pg. 19.



Traditional Approaches

The difference of two lattice vectors maybe smaller than the operands.

- Classical approaches such as LLL or BKZ subtract $b_i \in \mathcal{B}$ among themselves till certain conditions are met.
- Sieving algorithm for $P \subset \mathcal{L}(\mathcal{B})$.

  do

  $R \leftarrow \emptyset$

  for each $(u, v) \in P^2$

  $t \leftarrow v - u$

  if $(\vec{0} \neq t \notin P) \wedge (\|t\| < \|u\| \vee \|t\| < \|v\|)$ **then** $R \leftarrow R \cup \{t\}$

  $P \leftarrow \textbf{Select}(P \cup R)$

  while $R \neq \emptyset$
- Laarhoven [1] treat sieving as a GA by representing $v = \mathcal{B}x$ with $x \in \mathbb{Z}^d$.

  Pros Easy mutation through perturbed x_i.

  Cons Evaluating fitness (inverse norm) now requires a matrix multiplication, namely $\mathcal{B}x$.

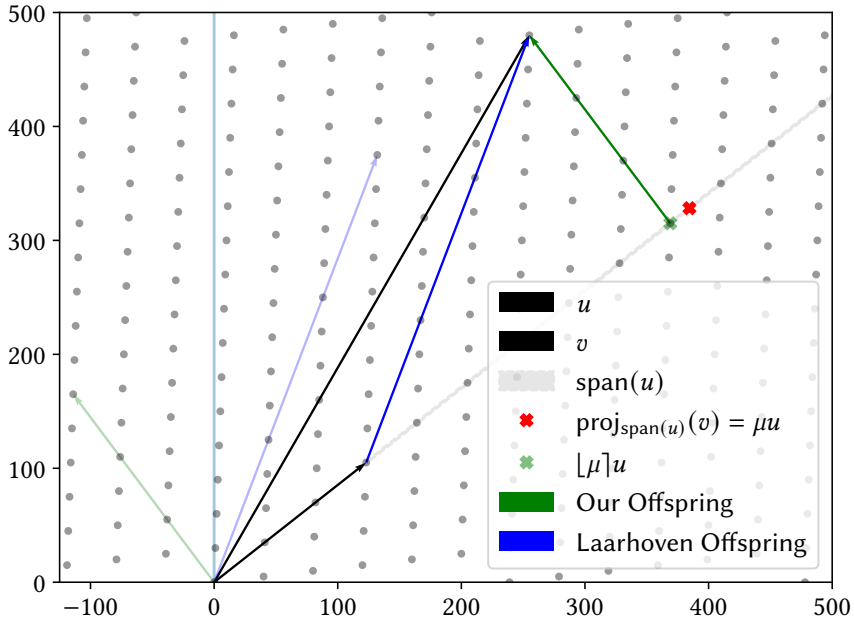


Our Domain-Informed Genetic Algorithm

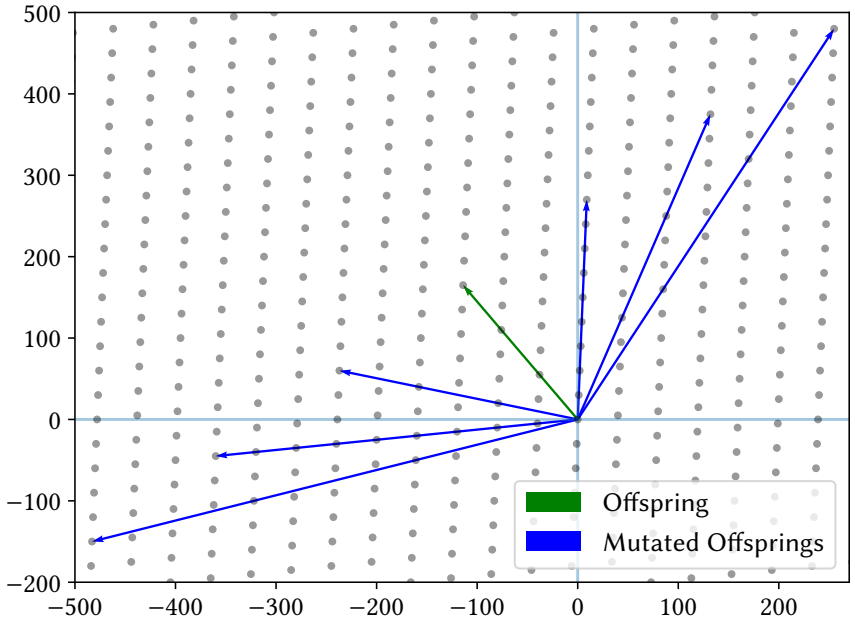
GA based on insights from classical methods.

We represent $v \in \mathcal{L}(\mathcal{B})$ as is and,

$$\textbf{Cross}(u, v) = v - \lfloor \mu \rceil \, u \quad \text{where} \quad \mu = \frac{\Re(u \cdot v)}{u \cdot u} + \frac{\Im(u \cdot v)}{u \cdot u} i.$$



(A) Illustration of our crossover technique.



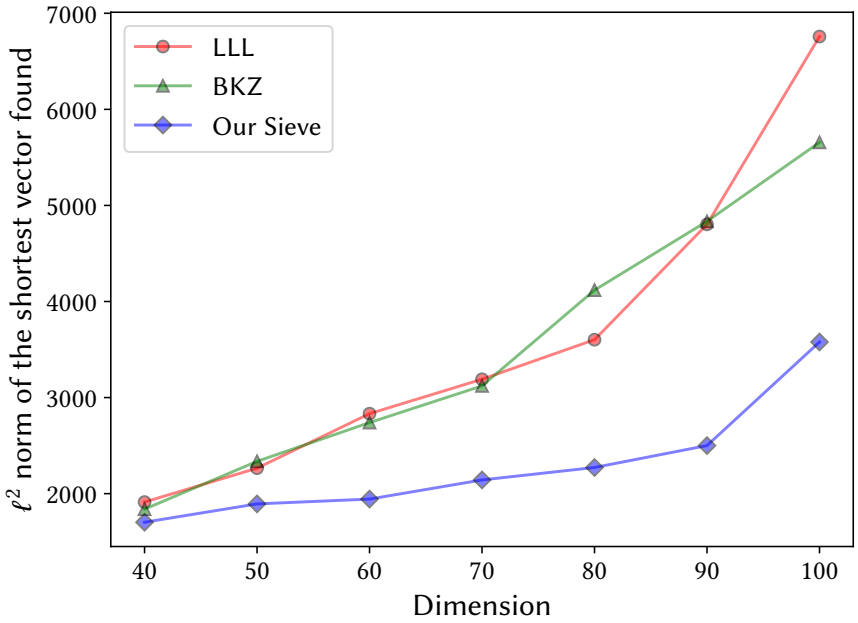
(B) Fig. 3a offspring mutations.

Figure: Illustration of our crossover and mutation techniques.

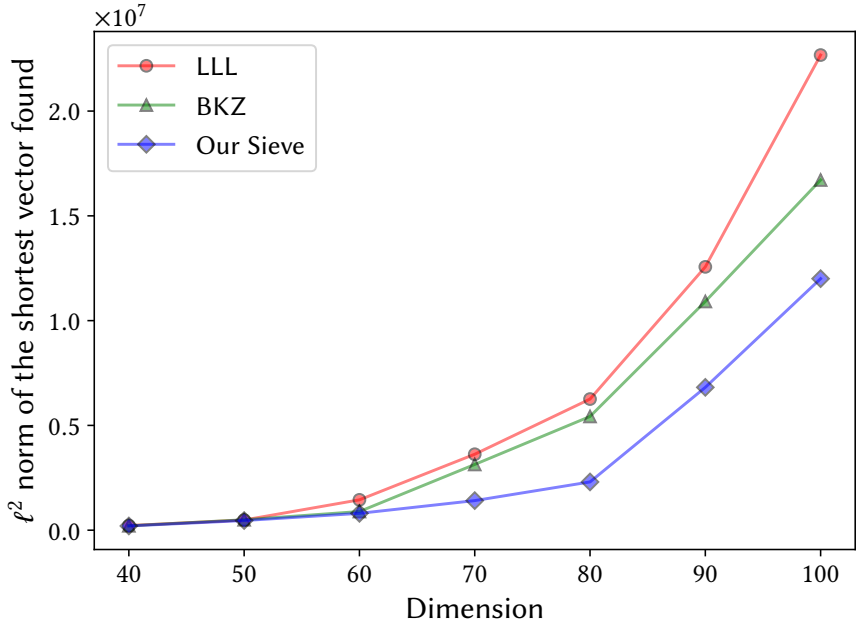
Test Data

Darmstadt Challenge, Random Integral & Module Lattices

We tested our GA on Darmstadt Lattice Challenges, randomly generated integral lattices and finally module lattices.



(A) Shortest vector length comparison in the SVP challenge lattices.



(B) Shortest vector length comparison in the randomly generated lattices.

Figure: Comparison of the shortest vector length found by different algorithms.

Reduction Performance

The α approximation of the shortest vector.

- Lattice cryptography relies on the inability to solve the apprSVP in high dimensions ($d > 300$) with an approximation factor $\alpha = \text{poly}(d)$.
- For all integral lattices of $d \leq 100$, we solved the apprSVP with $\alpha < 1.5$.
- For all module lattices of $d \leq 50$, we solved the apprSVP with $\alpha < 2.05$.
- Laarhoven solved the same $d = 40$ lattice as us. They reported 29 iterations starting with a set of 1500 lattice vectors to find the shortest vector of norm 1702 and an average vector of norm 2008.
- Our domain-informed crossover took 7 iterations starting with a set of 1000 lattice vectors to find the same shortest vector and an average vector of norm 1981.



Summary & Outlook

We optimised the GA approach to the SVP using domain specific knowledge.

- SVP is a popular choice for the quantum-resistant cryptographic algorithms.
- Three of NIST's selections for standard quantum-resistant algorithms are based on SVP [3].
- SVP is to find the shortest vector or an approximation thereof in an integral span of a $\mathcal{B} \in \mathbb{R}^{d \times d}$ called a Lattice.
- Traditional algorithms solve the SVP by reducing the entire basis.
- Laarhoven treat sieving for the shortest vector as a GA.
- We improve and speed-up the GA with a domain-informed crossover and representation for the SVP [2].
- Future research could tackle the SVP in higher dimensions $d > 100$ by different population management or selection methods.



Summary & Outlook

En garde!

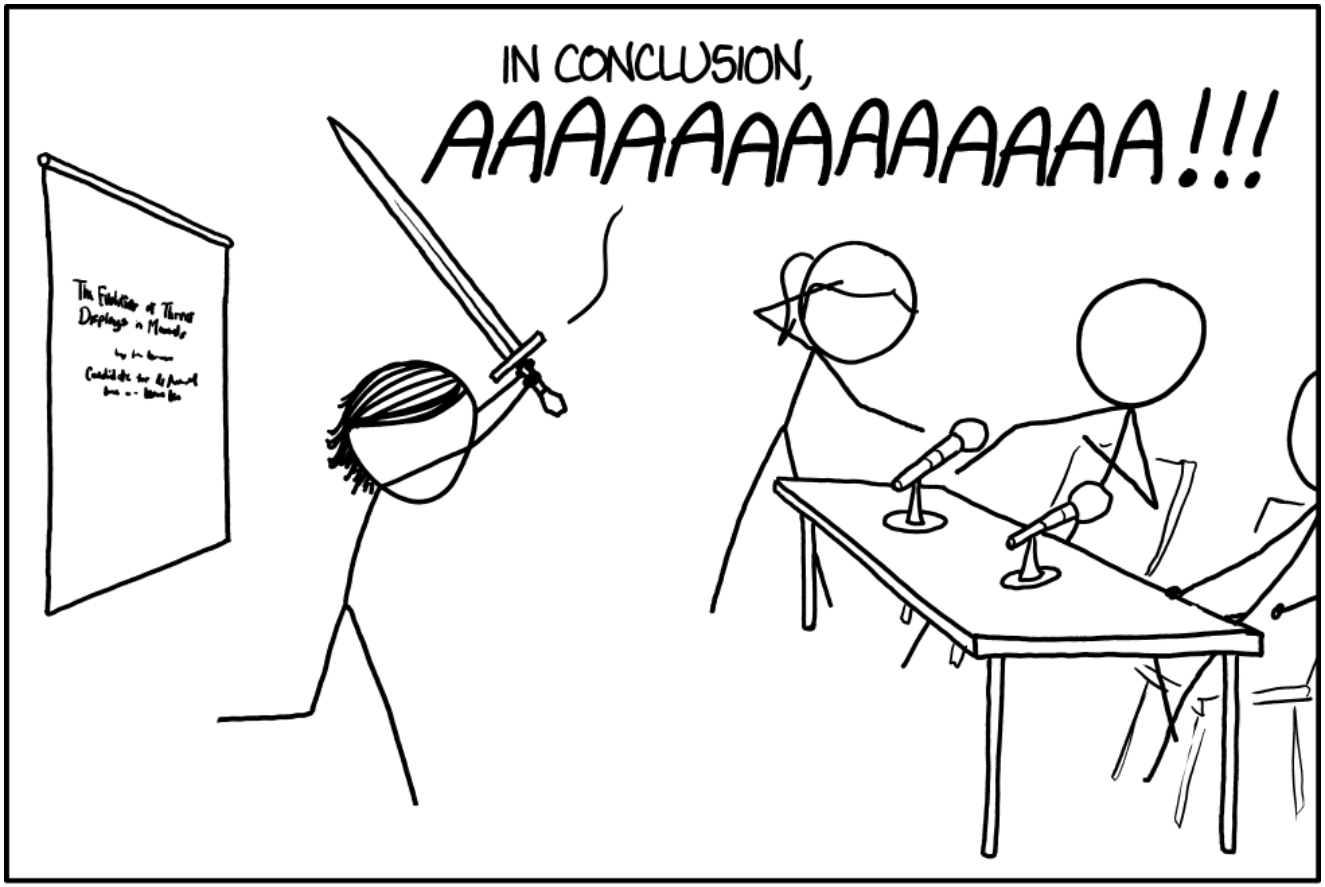


Figure: https://xkcd.com/1403/



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